Chapter 6.2 — Formula Sheet

Optimal Stopping Theory: the Secretary Problem

Companion to your Notebook 6 syllabus, your *20 Worked Examples* (Ex 1–20) and the *Stage-1 Hard Drill* (Q1–Q10)

A. Setup and Success Probability

The Rule and Its Win Probability [Ex 1–5; Q1]

n candidates, random order, relative ranks only, no recall, win only on the overall best. Skip s, then hire the first record:

$$\mathbb{P}_n(s) = \frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i} \quad (s \geq 1), \qquad \mathbb{P}_n(0) = \frac{1}{n}$$

Derivation: best at j ($\frac{1}{n}$) is hired iff the best of the first $j-1$ lies in the first s ($\frac{s}{j-1}$). Small n: $\frac{1}{2}$ (n=3), $\frac{11}{24}$ (n=4), $\frac{13}{30}$ (n=5).

Record Lemma [Ex 5; Q10]

Candidate j is a record w.p. $\frac{1}{j}$, *independently*:

$$\mathbb{E}[\#\text{records}] = H_n, \quad \mathbb{P}(\text{no record in } s{+}1..n) = \frac{s}{n}$$

$$\mathbb{E}[\#\text{records after } s] = H_n - H_s \xrightarrow{n\to\infty} \ln\tfrac{n}{s}$$

B. The $1/e$ Rule

Continuous Limit [Ex 6–8; Q3]

$x = s/n$, $\sum \frac{1}{i} \to -\ln x$:

$$\mathbb{P} \to -x\ln x, \qquad x^\star = \tfrac{1}{e}, \qquad \mathbb{P}^\star = \tfrac{1}{e} \approx 0.368$$

Flat top: $\phi(\frac{1}{4}) = \phi(\frac{1}{2}) = \frac{\ln 2}{2} = 0.347$. $\phi \geq 0.35 \iff x \in [0.259, 0.488]$. Skipping half at n=100 costs only 0.022.

Exact Optima [Ex 4, 7, 16; Q1, Q10]

n	10	20	30	50	100	1000
$s^\star$	3	7	11	18	37	368
$\mathbb{P}$	.399	.384	.379	.374	.371	.368

n=12: $s^\star$=4, $\mathbb{P} = \frac{32891}{83160} = 0.3955$. Use $s = \text{round}(n/e)$; approach to $\frac{1}{e}$ is from above.

DP Threshold: Stop at a Record [Ex 10–11; Q1, Q10]

Record at j: stop wins $\frac{j}{n}$, continue wins $\frac{j}{n}\sum_{i=j}^{n-1}\frac{1}{i}$. Stop iff

$$\sum_{i=j}^{n-1}\frac{1}{i} \leq 1$$

First such $j = r$; skip $s = r-1$. Large n: $\ln\frac{n}{r} = 1 \Rightarrow r = \frac{n}{e}$.

C. Outcome Laws and Robustness

Win / No Hire / Wrong Hire [Ex 9, 14; Q1, Q10]

$$\mathbb{P}(\text{no hire}) = \tfrac{s}{n} \to \tfrac{1}{e}, \quad \mathbb{P}(\text{win}) \to \tfrac{1}{e}, \quad \mathbb{P}(\text{wrong}) \to 1 - \tfrac{2}{e} = 0.264$$

$$\mathbb{P}(\text{win} \mid \text{hired}) = \frac{\mathbb{P}}{1-s/n} \to \frac{1}{e-1} = 0.582$$

Stopping Time [Ex 12; Q2, Q10]

$$\mathbb{P}(T=j) = \frac{s}{j(j-1)} \ (j>s), \qquad \mathbb{P}(\text{win}) = \sum_{j>s}\mathbb{P}(T=j)\frac{j}{n}$$

$$\mathbb{E}[T] = n\,\mathbb{P}_n(s) + s \ \to \ \tfrac{2n}{e} \approx 0.736\,n$$

n=100, s=37: 74.1. n=30, s=11: 22.36.

If n Is Uncertain [Ex 13]

A fixed *count* is fragile (s=37: 0.225 at n=50, 0.221 at n=400); a fixed *fraction* n/e stays ≈ 0.37. Observe for a fraction of elapsed time.

D. Odds Theorem

Bruss's Odds Theorem [Ex 17–18; Q4, Q5]

Independent events I_k, probabilities p_k, stop on the *last* success. Odds $r_k = \frac{p_k}{1-p_k}$. Start at the largest s with $\sum_{k\geq s} r_k \geq 1$ (s=1 if none), stop at the first success from s:

$$\mathbb{P}(\text{win}) = \prod_{k\geq s}(1-p_k)\cdot\sum_{k\geq s} r_k$$

Secretary: $p_k = \frac{1}{k}$, $r_k = \frac{1}{k-1}$. If the odds sum is *exactly* 1, one step earlier ties.

Dice and Uneven Signals [Q4, Q5]

Last six in 10 rolls: $r = \frac{1}{5}$, start roll 6: $\mathbb{P} = (\frac{5}{6})^5 = \frac{3125}{7776} = 0.402$. Last $\{5,6\}$: start roll 9, $\mathbb{P} = \frac{4}{9}$. Ten signals (Q5): start day 8, $\mathbb{P} = \frac{113}{250}$.

Poisson Arrivals, Last One [Q6]

Rate λ on $[0,T]$: act on the first arrival after $T-u$; win iff exactly one arrival in the window:

$$\mathbb{P} = \lambda u\, e^{-\lambda u}, \quad u^\star = \tfrac{1}{\lambda}, \quad \mathbb{P}^\star = \tfrac{1}{e}$$

If $\lambda T < 1$: start at once, $\mathbb{P} = \lambda T\, e^{-\lambda T}$ (0.354 for λT=0.75).

E. Variants

Full Information (Iid $U(0,1)$, Pick the Max) [Ex 19; Q7]

Accept a record x with k draws left iff $x^k \geq$ continuation value.

n=2 : $\frac{1}{2} \to \frac{3}{4}$ $\qquad$ n=3 : $t = \frac{1+\sqrt{6}}{5}$, $\frac{1}{2} \to \frac{293}{600} + \frac{2\sqrt{6}}{25} = 0.6843$

n=10: 0.609. Limit 0.580, versus 0.368 with ranks only.

Other Objectives: DP on (Position, Relative Rank) [Q8, Q9]

State (j, r); $C_j = \frac{1}{j+1}\sum_r \max(\text{accept}_{j+1,r}, C_{j+1})$. Chance that relative rank r at j has absolute rank k:

$$\frac{\binom{k-1}{r-1}\binom{n-k}{j-r}}{\binom{n}{j}}$$

Best or second best (n=4): $\frac{3}{4}$ vs $\frac{11}{24}$ for best only. **Min expected rank**: $\mathbb{E}[\text{rank} \mid j, r] = \frac{r(n+1)}{j+1}$; n=4: $\frac{15}{8}$, n=10: $\frac{3223}{1260}$, limit ≈ 3.87.

F. Capstone Recipe

Any Secretary Question in 60 Seconds [Ex 20; Q10]

1. Cut-off: smallest r with $\sum_{i=r}^{n-1}\frac{1}{i} \leq 1$, skip $r-1 \approx \frac{n}{e}$. 2. $\mathbb{P} = \frac{s}{n}\sum_{i=s}^{n-1}\frac{1}{i}$; no hire $= \frac{s}{n}$; wrong $= 1 - \mathbb{P} - \frac{s}{n}$. 3. $\mathbb{E}[T] = n\mathbb{P} + s$. 4. Record at j: compare $\frac{j}{n}$ with $\frac{j}{n}\sum_{i=j}^{n-1}\frac{1}{i}$. n=20: s=7, $\mathbb{P} = 0.384$, $\mathbb{E}[T] = 14.68$.

Reminder: "best so far" is a record w.p. $\frac{1}{j}$; the cut-off is where the expected number of future records hits one; the odds theorem handles any independent signals; always check a stopping rule by simulation.